



PHP Practicum

IS4102

Presentation 02

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Content

- ▢ Scope of Variables
- ▢ Local Variables
- ▢ Global Variables
- ▢ Static Variables
- ▢ Function Parameters
- ▢ Operators
- ▢ Arithmetic Operators
- ▢ Comparison Operators
- ▢ Logical Operators



Scope of Variables

□ *The range of availability*

Local
Variables

Function
Parameters

Global
Variables

Static
Variables

Local Variables

- Can be declared in a function
- Assigned value for a variable in a function is different from the outside.

```
<?php
```

```
    $x = 10;
```

```
    function assignAnotherValue(){
```

```
        $x = 15;
```

```
        echo "Inside value of \$x is $x <br>";
```

```
    }
```

```
    assignAnotherValue();
```

```
    echo "Outside value of \$x is $x";
```

```
?>
```

Inside value of \$x is 15
Outside value of \$x is 10

Global Variables

- Can Be accessed in any part of the program.
- Must be explicitly declared to be global in the function in which it is to be modified.
- Uses the keyword **"GLOBAL"**.

```
<?php
    $x = 10;

    function testGlobalValue(){
        GLOBAL $x;
        $x++;
        echo "Value of \$x is $x <br>";
    }

    testGlobalValue();
    echo "Outside value of \$x is $x";
?>
```

Value of \$x is 11
Outside value of \$x is 11

Static Variables

- ❑ Declared in a function
- ❑ Will not lose its value after the end of function
- ❑ Will hold assigned value until for next call of the same function
- ❑ Uses the keyword **"STATIC"**

```
<?php
function testStaticVariable(){
    STATIC $x = 5;
    $x++;
    echo "Now the value of \$x is $x <br>";
}

testStaticVariable();
testStaticVariable();
?>
```

Now the value of \$x is 6
Now the value of \$x is 7

Function Parameters

- ▮ Declared after the function name and inside parentheses

```
<?php
    function calculateArea($height, $width){
        $area = $height*$width;
        return $area;
    }

    $getArea = calculateArea(2,5);
    echo "The area is $getArea";
?>
```

The area is 10



Operators

- ✓ Arithmetic
- ✓ Comparison
- ✓ Logical/Relational
- ✓ Assignment
- ✓ Conditional/Ternary

Arithmetic Operators

Operator	Description
+	<i>Add two operands</i>
-	<i>Subtracts second operand from the first</i>
*	<i>Multiply both operands</i>
/	<i>Divide the numerator by denominator</i>
%	<i>Modulus Operator and remainder of after an integer division</i>
++	<i>Increment operator, increases integer value by one</i>
--	<i>Decrement operator, decreases integer value by one</i>

Arithmetic Operators (cont.)

□ Exercise:

Write PHP syntax to display;

- *Addition*
- *Subtraction*
- *Multiplication*
- *Division*
- *Modulus*
- *Increment*
- *Decrement*

```
<?php
    $x=35;
    $y=6;

    echo $x+$y;
    echo "<br>";echo "<br>";
    echo $x-$y;
    echo "<br>";echo "<br>";
    echo $x*$y;
    echo "<br>";echo "<br>";
    echo $x/$y;
    echo "<br>";echo "<br>";
    echo $x%$y;
    echo "<br>";echo "<br>";
    echo $x++;
    echo "<br>";echo "<br>";
    echo ++$x;
    echo "<br>";echo "<br>";
    echo $y--;
    echo "<br>";echo "<br>";
    echo --$y;
    echo "<br>";echo "<br>";
```

?>

41

29

210

5.833333333333333

5

35

37

6

4



Exercise 01

- Write a PHP script that calculates the area and perimeter of a rectangle given its length and width?

```
<?php
    $length = 10;
    $width = 5;

    $area = $length * $width;
    $perimeter = 2 * ($length + $width);

    echo "Area: $area\n";
    echo "Perimeter: $perimeter\n";
?>
```



Answer 01

```
<?php
```

```
    $length = 10;
```

```
    $width = 5;
```

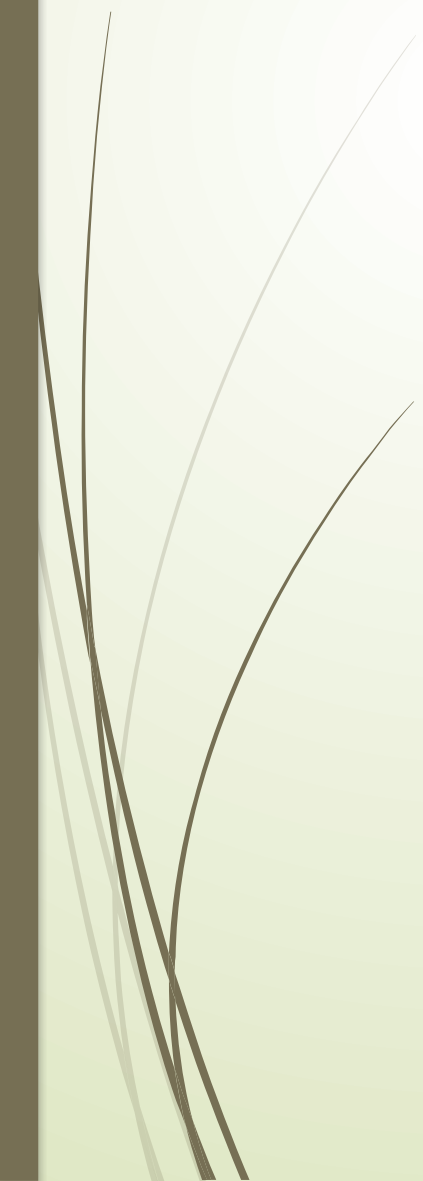
```
    $area = $length * $width;
```

```
    $perimeter = 2 * ($length + $width);
```

```
    echo "Area: $area\n";
```

```
    echo "Perimeter: $perimeter\n";
```

```
?>
```



Comparison Operators

Operator	Description
<code>==</code>	<i>Check if the value of two operands are equal or not</i>
<code>!=</code>	<i>Check if the value of two operands are equal or not</i>
<code>></code>	<i>Check if the value of left operand is greater than the value of right operand</i>
<code><</code>	<i>Check if the value of left operand is less than the value of right operand</i>
<code>>=</code>	<i>Check if the value of left operand is greater than or equal to the value of right operand</i>
<code><=</code>	<i>Check if the value of left operand is greater than or equal to the value of right operand</i>

Comparison Operators (cont.)

```
<?php
if ($a > $b) {
    echo "a is bigger than b";
}
elseif ($a == $b) {
    echo "a is equal to b";
}
else {
    echo "a is smaller than b";
}
?>
```

Comparison Operators (cont.)

```
if ($a > $b){  
    echo $a." is greater than ".$b;  
}  
elseif ($a == $b){  
    echo $a." equals ".$b;  
}  
else{  
    echo $a." is neither greater than or equal to ".$b;  
}  
endif;  
?>
```

Comparison Operators (cont.)

Exercise 02:

Write PHP syntax to display the results of comparison operators

You may need syntax structure of Decision making

```
If(condition){  
    //occurs if TRUE  
}  
else{  
    //occurs if FALSE  
}
```

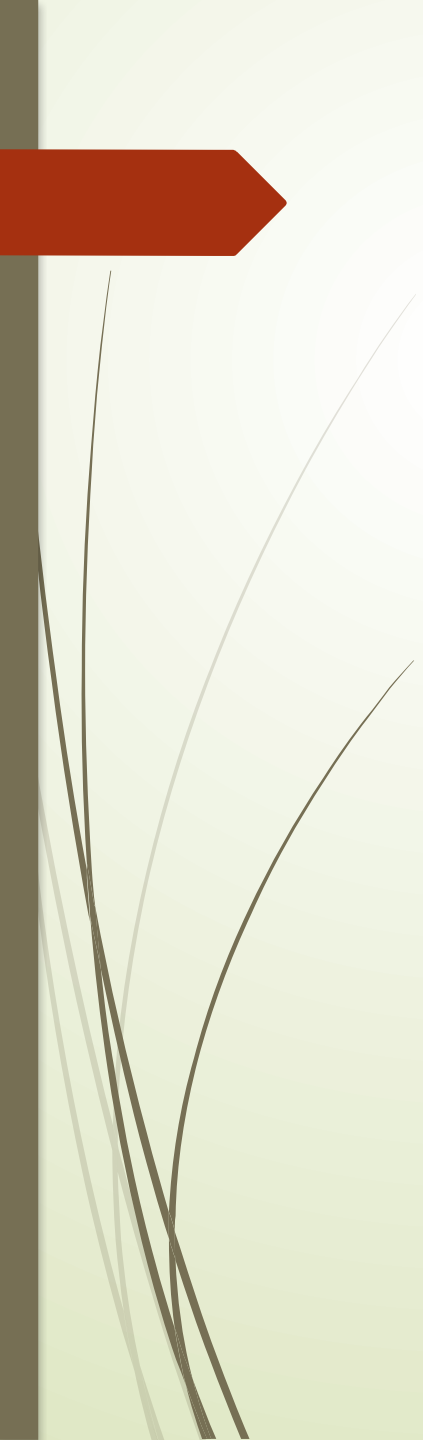


Answer 02

```
<?php
$a = 10;
$b = 5;

echo "a = $a, b = $b\n\n";

// Equal (==)
if ($a == $b) {
    echo "$a is equal to $b\n";
} else {
    echo "$a is NOT equal to $b\n";
}
```



```
// Not Equal (!=)
```

```
if ($a != $b) {
```

```
    echo "$a is NOT equal to $b\n";
```

```
} else {
```

```
    echo "$a is equal to $b\n";
```

```
}
```

```
// Greater Than (>)
```

```
if ($a > $b) {
```

```
    echo "$a is greater than $b\n";
```

```
} else {
```

```
    echo "$a is NOT greater than $b\n";
```

```
}
```

```
// Less Than (<)
```

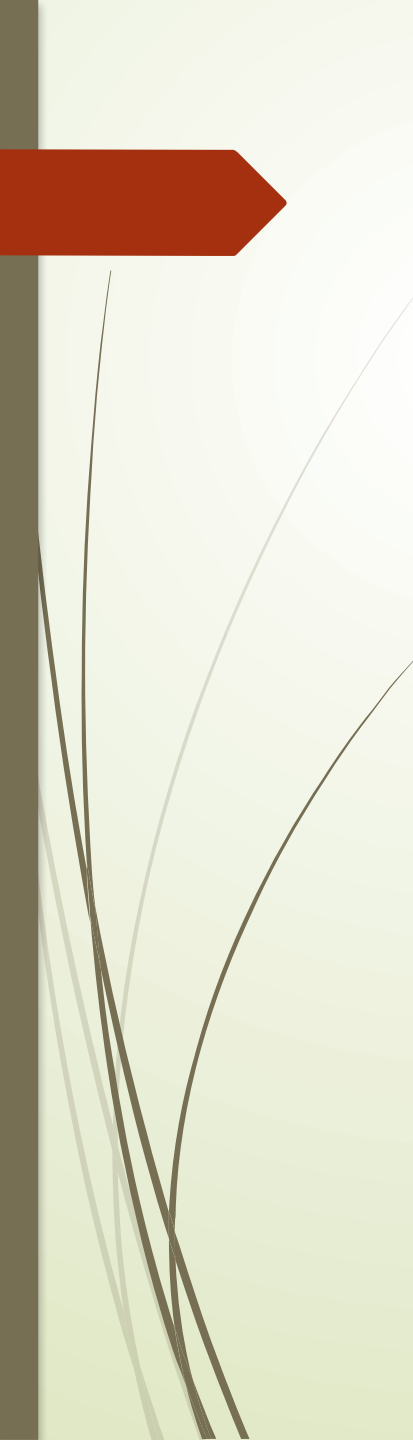
```
if ($a < $b) {
```

```
    echo "$a is less than $b\n";
```

```
} else {
```

```
    echo "$a is NOT less than $b\n";
```

```
}
```



```
// Greater Than or Equal To (>=)
```

```
if ($a >= $b) {
```

```
    echo "$a is greater than or equal to $b\n";
```

```
} else {
```

```
    echo "$a is NOT greater than or equal to $b\n";
```

```
}
```

```
// Less Than or Equal To (<=)
```

```
if ($a <= $b) {
```

```
    echo "$a is less than or equal to $b\n";
```

```
} else {
```

```
    echo "$a is NOT less than or equal to $b\n";
```

```
}
```

```
// Identical (===)
```

```
if ($a === $b) {
```

```
    echo "$a is identical to $b (same value & type)\n";
```

```
} else {
```

```
    echo "$a is NOT identical to $b (either value or type is different)\n";
```

```
}
```



```
// Not Identical (!==)
```

```
if ($a !== $b) {
```

```
    echo "$a is NOT identical to $b (either value or type is different)\n";
```

```
} else {
```

```
    echo "$a is identical to $b (same value & type)\n";
```

```
}
```

```
?>
```

Logical Operators

Operator	Description
<i>and , &&</i>	<i>True if both operands are true or non zero</i>
<i>or , </i>	<i>True if either one of operands is true or non zero</i>
<i>!</i>	<i>Reverse the logical state</i>

Exercise 03: *Write PHP syntax to display the results of Logical operators*

Answer 03

```
<?php
```

```
    $a = true;
```

```
    $b = false;
```

```
    echo "a = " . ($a ? "true" : "false") . "\n";
```

```
    echo "b = " . ($b ? "true" : "false") . "\n\n";
```

```
    echo "AND (&&): " . (($a && $b) ? "true" : "false") . "\n"; // False (true  
AND false)
```

```
    echo "OR (||): " . (($a || $b) ? "true" : "false") . "\n"; // True (true OR  
false)
```

```
    echo "NOT (!a): " . (! $a) ? "true" : "false") . "\n"; // False (NOT true)
```

```
    echo "XOR (a xor b): " . (($a xor $b) ? "true" : "false") . "\n"; // True  
(true XOR false)
```

```
?>
```



Thank you!